

ABERDARE TECHNICAL COLLEGE

**UNIT OF COMPETENCY: DESIGN COMPUTERIZED
DATABASE**

LEVEL: 5

UNIT CODE: 0612 451 08A

ELEMENT COVERED: DESIGN COMPUTERIZED DATABASE

CAT 1, TERM 3

TIME: 1 HOUR 30 MINUTES

TOTAL MARKS: 50

Instructions to Candidates

- ✓ *Write your name and admission number in the space provided.*
- ✓ *This paper has **THREE** sections: A, B, and C. Answer **ALL** questions in all sections.*
- ✓ *Write your answers legibly in the spaces/answer booklet provided.*
- ✓ *Do not write on the question paper.*

Name: _____ Admission no.....Date:

.....sign.....

SECTION A: (20 Marks)

1. Define the following terms as used in database management systems: (4 marks) a) Database b) Database Management System (DBMS) c) Data redundancy d) Data independence
2. State **four** functions of a database management system. (4 marks)
3. Outline **four** application areas of database systems in real life, giving a brief example for each. (4 marks)
4. Distinguish between the following types of database management systems, giving one example of each: (4 marks) a) Centralized DBMS b) Distributed DBMS
5. Explain **two** advantages of using a DBMS over a traditional file-based system. (4 marks)

SECTION B: (20 Marks)

6. With the aid of a well-labelled diagram, describe the **five** major components of a DBMS environment. (10 marks)
7. Explain the **three** tier database architecture(6 marks)
8. a) Define the term **requirements elicitation** as used in database design. (2 marks)

b) State **two** techniques a database analyst can use to gather database requirements from users. (2 marks)

SECTION C: (10 Marks)

9. a) Outline **three** components that should be included in a database requirement assessment report. (3 marks)

b) Compare the **relational database model** and the **hierarchical database model** under the following aspects: (7 marks)
 - i. Structure of data representation (2 marks)
 - ii. Relationship between records/entities (2 marks)
 - iii. Flexibility in querying data (2 marks)

iv. One real-world example of a system that uses each model (1 mark)